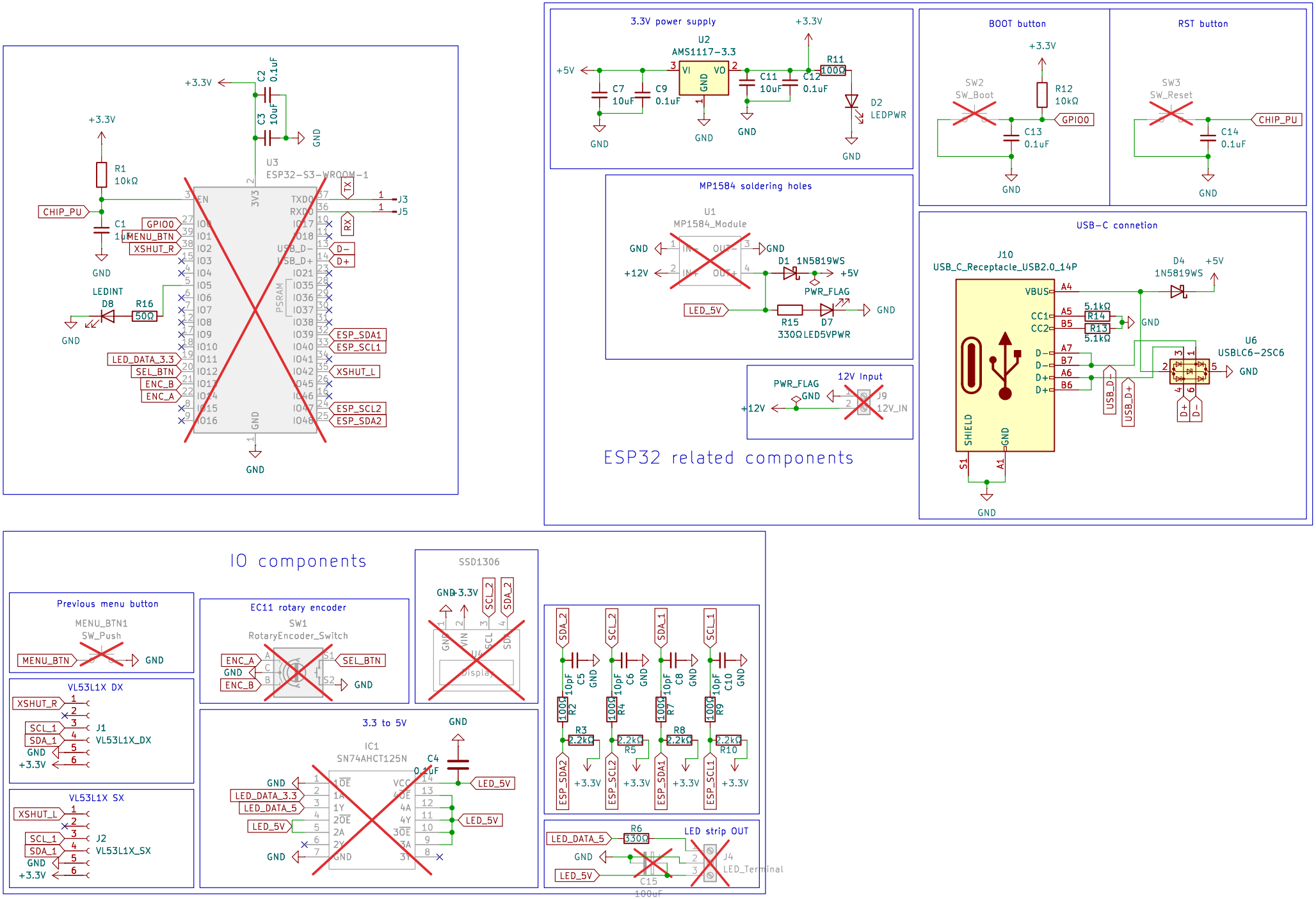
3.3V power supply
U2
AMS1117-3.3
+5V
+3.3V
C7 10uF
C9 0.1uF
C11 10uF
C12 0.1uF
R11 100Ω
D2 LEDPWR
GND
BOOT button
SW2 SW_Boot
R12 10kΩ
C13 0.1uF
GPIO0
RST button
SW3 SW_Reset
C14 0.1uF
CHIP_PU
MP1584 soldering holes
U1
MP1584_Module
IN- OUT-
IN+ OUT+
+12V
D1 1N5819WS
PWR_FLAG
LED_5V
R15 330Ω
D7 LED5VPWR
12V Input
PWR_FLAG
J9 12V_IN
ESP32 related components
USB-C connetion
J10
USB_C_Receptacle_USB2.0_14P
VBUS
CC1
CC2
D-
D+
SHIELD
S1
A1
A4
A5
B5
A7
B7
A6
B6
5.1kΩ R14
5.1kΩ R13
D4 1N5819WS
U6
USBLC6-2SC6
USB_D-
USB_D+
C2 0.1uF
C3 10uF
R1 10kΩ
C1 1uF
U3
ESP32-S3-WROOM-1
EN
3V3
TXD0
RXD0
IO17
IO18
USB_D-
USB_D+
IO21
PSRAM
IO35
IO36
IO37
IO38
IO39
IO40
IO41
IO42
IO45
IO46
IO47
IO48
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO11
IO12
IO13
IO14
IO15
IO16
GPIO0
MENU_BTN
XSHUT_R
LEDINT
D8
R16 50Ω
LED_DATA_3.3
SEL_BTN
ENC_B
ENC_A
ESP_SDA1
ESP_SCL1
XSHUT_L
ESP_SCL2
ESP_SDA2
TX
RX
J3
J5
IO10 IO11 IO12 IO13 IO14 IO15 IO16
IO Components
Previous menu button
MENU_BTN1
SW_Push
EC11 rotary encoder
SW1
RotaryEncoder_Switch
ENC_A
ENC_B
SEL_BTN
SSD1306
GND
+3.3V
SCL_2
SDA_2
GND
VIN
SCL
SDA
U4
Display
VL53L1X DX
XSHUT_R
SCL_1
SDA_1
J1
VL53L1X_DX
+3.3V
VL53L1X SX
XSHUT_L
J2
VL53L1X_SX
3.3 to 5V
IC1
SN74AHCT125N
C4 0.1uF
LED_DATA_3.3
LED_DATA_5
LED_5V
1OE
1A
1Y
2OE
2A
2Y
GND
VCC
4OE
4A
4Y
3OE
3A
3Y
SDA_2
SCL_2
SDA_1
SCL_1
C5 10pF
C6 10pF
C8 10pF
C10 10pF
R2 100Ω
R4 100Ω
R7 100Ω
R9 100Ω
R3 2.2kΩ
R5 2.2kΩ
R8 2.2kΩ
R10 2.2kΩ
ESP_SDA2
ESP_SCL2
ESP_SDA1
ESP_SCL1
R6 330Ω
LED strip OUT
J4
LED_Terminal
C15 100uF